Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Binomial Distribution

SOLUTIONS

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1. Fixed n; two outcomes; independent trials; constant p.
2. $\binom{5}{2}(0.4)^2(0.6)^3 = 0.2304$. **3.** $(0.7)^6 \approx 0.1176$.
4. $\binom{8}{3}(0.25)^3(0.75)^5 \approx 0.2076$. **5.** Sum $r = 0, 1, 2$.
6. $E(X) = 2.4$; Var $= 0.96$. **7.** Mean 4.8; SD $\sqrt{2.88} \approx 1.70$.
8. $\binom{8}{5}(\frac{1}{2})^8 = \frac{56}{256} = \frac{7}{32}$. **9.** $\binom{5}{2}(\frac{1}{6})^2(\frac{5}{6})^3$.
10. $1 - (0.8)^7$. **11.** $\binom{9}{4}(0.35)^4(0.65)^5$.
12. $\binom{15}{2}(0.1)^2(0.9)^{13}$. **13.** $\binom{20}{5}(0.3)^5(0.7)^{15}$.
14. $np = 12$, $np(1-p) = 3 \Rightarrow p = \frac{3}{4}$, $n = 16$. **15.** Mode $= 2$ or 3.
16. Sum $P(X = 3) + \cdots + P(X = 6)$. **17.** $P(X = 10) + P(X = 11) + P(X = 12)$.
18. Cumulative or sum. **19.** $E = 17$; $P = (0.85)^{20}$.
20. $n = 20$; $P(X = 2) = \binom{20}{2}(0.25)^2(0.75)^{18}$.
21. $\binom{10}{3}(\frac{1}{4})^3(\frac{3}{4})^7$. **22.** $(1-p)^8 = 0.01 \Rightarrow p = 1 - 0.01^{1/8}$; $E = np$.
23. Accidents not independent / p not constant / n not fixed.
24. Find largest a with $P(X \le a) \le 0.1$; smallest b with $P(X \ge b) \le 0.1$.
25. $X + Y \sim B(10, 0.4)$; $P(X + Y = 3) = \binom{10}{3}(0.4)^3(0.6)^7$.